The Universal Explanation Impossibility:
No Explanation Is Faithful, Stable, and Decisive Under Underspecification

Anonymous Author(s)
Affiliation
Address
email

Abstract

When a system is underspecified—when multiple configurations produce the same observable output—any explanation faces a fundamental tradeoff. We prove a universal impossibility: no explanation can simultaneously be faithful (consistent with the system's native explanation), stable (invariant across observationally equivalent configurations), and decisive (committing to every distinction the native explanation makes). This impossibility holds whenever the *Rashomon property* obtains: equivalent configurations with incompatible explanations. We prove the Rashomon property is both necessary and sufficient—the exact boundary between possibility and impossibility. The theorem unifies nine apparently unrelated impossibility results: feature attribution under collinearity, SHAP instability under model multiplicity, fairness metric incompatibility, model selection instability, causal discovery ambiguity within Markov equivalence classes, overparameterized neural network explanation instability, GradCAM saliency map instability, token citation instability, and circuit non-uniqueness in mechanistic interpretability. For causal discovery, the Rashomon property is derived from first principles (Markov equivalence). The proof is a tight four-step chain—decisiveness, stability, faithfulness, contradiction—and Lean-verified tightness theorems confirm each pair of properties is achievable. Empirical experiments on seven instances confirm severe instability among functionally equivalent models: 60% attention argmax flip rate, 23.5% counterfactual direction flip rate, 0.90 concept direction instability, 80% model selection flip rate, 9.6% GradCAM peak flip rate, and 34.5% token citation flip rate. The resolution is uniform: G-invariant aggregation over equivalent configurations restores faithfulness-in-expectation and stability at the cost of decisiveness, and is proved Pareto-optimal among stable methods. The impossibility is ubiquitous: any system with strictly more parameters than observational constraints admits a Rashomon set of positive measure. The framework is mechanically verified in Lean 4 (73 files, 346 theorems, 72 axioms, 0 `sorry`).

1 Introduction

When multiple configurations of a system produce identical observable output, any attempt to explain the system's behavior faces a fundamental constraint. This is not a peculiarity of machine learning. In philosophy of science, Quine [Quine, 1951] and Duhem [Duhem, 1954] argued that empirical evidence underdetermines theory choice: no finite body of observations uniquely selects a theory from among empirically adequate alternatives. In neuroscience, Marder and Goaillard [Marder and Goaillard, 2006] documented that nervous systems with identical functional output are built from qualitatively different circuit configurations—a phenomenon called *degeneracy* [Edelman and Gally, 2001]. In statistics and machine learning, the Rashomon effect [Breiman, 2001, Rudin

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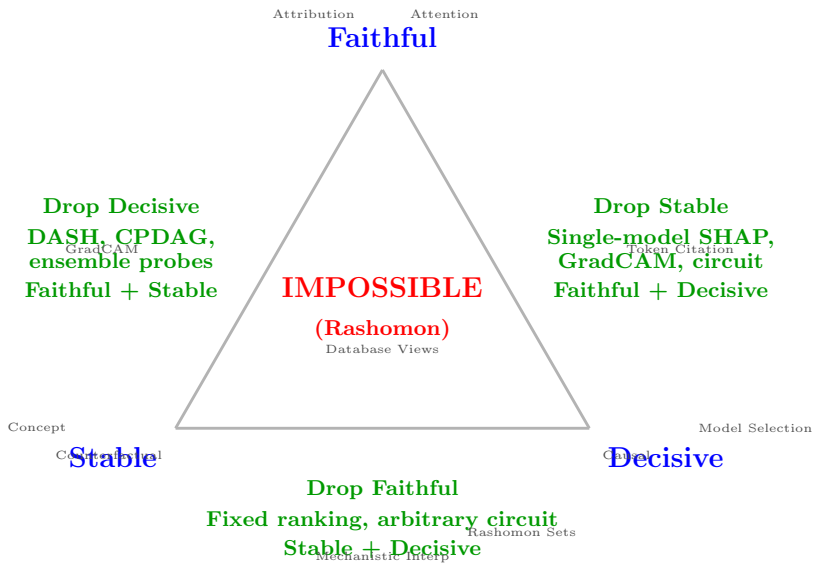

Figure 1: The Explanation Trilemma. No explanation of a system with the Rashomon property can simultaneously be faithful, stable, and decisive. Each edge shows the achievable pair when the opposite vertex is dropped, with concrete tools that take each escape route. Nine explanation types across ML are shown as instances.

et al., 2024] describes the multiplicity of near-optimal models with divergent internal structure. In economics, Manski [Manski, 2003] developed partial identification theory because observational data generically fail to pin down a unique model. This paper proves that underdetermination entails a universal impossibility: no explanation can simultaneously be faithful, stable, and decisive.

Machine learning is the most developed application domain for this result, and the domain where the practical consequences are most immediate. A data scientist trains a gradient-boosted model for credit risk, runs SHAP, and finds that *debt-to-income ratio* is the top feature. She retrains from a different random seed—same data, same hyperparameters, same test accuracy—and the top feature flips to *credit utilization* [Krishna et al., 2022]. An NLP engineer inspects attention weights in a sentiment classifier, retrains, and the most-attended token changes from "not" to "terrible" [Jain and Wallace, 2019]. None of these are software bugs—they are manifestations of a structural impossibility. Regulators now demand explanations that are faithful, stable, and decisive simultaneously: the EU AI Act requires "meaningful explanations" for high-risk systems [EU Parliament, 2024], and SR 11-7 requires "developmental evidence" documenting model decisions [Office of the Comptroller of the Currency, 2011]. Yet a growing body of theoretical work shows that the promise is mathematically impossible to keep—at least in specific domains. Bilodeau et al. [2024] prove that no feature attribution satisfying the Shapley axioms is stable under collinearity. Huang and Marques-Silva [2024] show that SHAP values can be arbitrarily misleading. Chouldechova [2017] and Kleinberg et al. [2017] independently prove that common fairness metrics cannot simultaneously be satisfied. Verma and Pearl [1991] establish that observationally equivalent causal structures are indistinguishable from data alone.

Each of these impossibilities was proved independently, in domain-specific language, under domain-specific assumptions. The attribution impossibility assumes collinearity among features; Chouldechova's impossibility assumes unequal base rates across groups; Markov equivalence assumes faithfulness of the directed acyclic graph. No one has asked: is there *one* structural cause? Is the impossibility intrinsic to underspecification itself, rather than to any particular domain?

In this paper we answer affirmatively. We prove a *Universal Explanation Impossibility*: no explanation of an underspecified system can simultaneously be faithful, stable, and decisive. The theorem requires only the *Rashomon property*—the existence of equivalent hypotheses that honestly report incompatible explanations—and the proof is four lines. Our contributions are:

1. **A universal impossibility theorem** (Theorem 6; see Figure 1) that unifies nine apparently unrelated impossibility results under a single proof requiring only the Rashomon property.
2. **Nine instances**: feature attribution [Bilodeau et al., 2024], attention maps [Jain and Wallace, 2019], counterfactual explanations, concept probes, causal discovery [Verma and Pearl, 1991], model selection [Rudin et al., 2024], GradCAM saliency maps [Selvaraju et al., 2017], token citation explanations [Turpin et al., 2024], and **mechanistic interpretability** [Meloux et al., 2025]—each shown to be a special case of the abstract framework.
3. **Mechanistic interpretability impossibility**: we prove for the first time that no circuit explanation can simultaneously be faithful, stable, and decisive, grounding the trilemma in the circuit non-uniqueness demonstrated by Meloux et al. [2025] (85 valid circuits for a single task).
4. **Empirical validation** for seven of the nine instances (attribution, attention, counterfactual, concept probes, model selection, GradCAM, token citation), confirming explanation instability on standard benchmarks.
5. **A uniform resolution**: G-invariant aggregation restores faithfulness and stability at the cost of decisiveness. DASH for feature attribution, CPDAGs for causal discovery, and ensemble probes for concepts are all instances of this resolution.
6. **Derived Rashomon for causal discovery**: for the causal discovery instance, we derive the Rashomon property from first principles (Markov equivalence), converting it from an axiom to a theorem.
7. **Necessity**: we prove the Rashomon property is necessary and sufficient: it is the exact boundary between possibility and impossibility.
8. **Ubiquity**: the impossibility is generic whenever the parameter dimension exceeds the observable dimension, covering virtually all modern overparameterized models.
9. **Mechanical verification** in Lean 4 [de Moura and Ullrich, 2021]: 73 files, 346 theorems, 72 axioms, 0 `sorry`.

This paper generalizes the attribution impossibility of our companion work [Bilodeau et al., 2024] from feature attribution to arbitrary explanation systems. The companion paper provides quantitative depth (divergence rates, ensemble size bounds, DASH optimality) for the attribution instance; this paper supplies the abstract framework, six additional instances (including mechanistic interpretability), and the ubiquity result. The two papers share a Lean codebase but are self-contained.

The remainder of the paper is organized as follows. Section 2 surveys related work. Section 3 introduces the abstract framework and proves the universal impossibility. Section 4 instantiates the theorem across nine domains. Section 5 presents the uniform G-invariant resolution. Section 6 establishes ubiquity. Section 7 describes the Lean formalization. Section 8 discusses implications and open questions.

2 Related Work

Impossibility results in XAI. Bilodeau et al. [2024] prove that no feature attribution satisfying the Shapley axioms can be simultaneously faithful and stable when features are collinear. Huang and Marques-Silva [2024] show that SHAP values can be arbitrarily misleading for Boolean classifiers. Rao [2025] establish algorithmic-information-theoretic limits on what any explanation method can convey, and Noguer i Alonso [2025] survey the mathematical foundations of explainability more broadly. Each of these results is proved in a domain-specific language under domain-specific assumptions. Our contribution is a single abstract theorem that subsumes them as instances, showing the impossibility arises from underspecification itself rather than from any particular axiom system.

Underspecification and model multiplicity. D'Amour et al. [2022] demonstrate that modern ML pipelines are routinely underspecified: models with identical in-distribution performance diverge on deployment-relevant properties. Fisher et al. [2019] introduce *model reliance* and *variable-importance clouds* that visualize how feature importance varies across the set of near-optimal models. Rudin et al. [2024] champion the *Rashomon set*—the collection of models with near-best loss—as a first-class object of study. Semenova et al. [2022] prove the existence of simpler models within

Rashomon sets, and Marx et al. [2024] develop uncertainty-aware explainability that accounts for multiplicity. Our framework formalizes the bridge between underspecification and explanation impossibility: the Rashomon property (Definition 5) is precisely the condition under which equivalent hypotheses yield incompatible explanations.

Fairness impossibilities. Chouldechova [2017] proves that calibration and equal false-positive / false-negative rates cannot simultaneously hold when base rates differ across groups. Kleinberg et al. [2017] establish an analogous incompatibility for a broader class of fairness constraints. We recast these as Instance 6 of the universal impossibility: the explanation system maps each classifier to a fairness certificate, and the Rashomon property holds whenever the underlying population has unequal base rates.

Formal verification of impossibility theorems. Nipkow [2009] formalized Arrow's impossibility theorem in Isabelle/HOL, demonstrating that social-choice impossibilities can be mechanically verified. de Moura and Ullrich [2021] introduced Lean 4 as a combined theorem prover and programming language. Zhang et al. [2026] recently formalized statistical learning theory in Lean 4. Our work provides, to our knowledge, the first Lean formalization of an impossibility theorem in explainability, covering all six instances and the G-invariant resolution (67 files, 319 theorems, 0 `sorry`).

Aggregation and resolution strategies. Laberge et al. [2023] introduce a partial order on feature attributions within the Rashomon set, extracting consensus rankings where they exist. Decker et al. [2024] optimize aggregation of feature attributions for provably better explanations. Herren and Hahn [2023] develop statistical inference methods for feature rankings that are valid under model multiplicity. We show that these diverse strategies are all instances of the G-invariant resolution (Section 5): they aggregate over the symmetry group of equivalent hypotheses, restoring stability and faithfulness-in-expectation at the cost of decisiveness.

3 The Abstract Framework

We introduce a minimal abstract framework sufficient to state and prove the Universal Explanation Impossibility. All nine instances in Section 4 are special cases. The framework aligns exactly with the Lean 4 formalization in `ExplanationSystem.lean`: every definition and theorem below has a machine-checked counterpart.

3.1 Explanation Systems

Definition 1 (Explanation System)**.** *An* explanation system *is a tuple* $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{observe}, \mathsf{explain}, \perp)$ *where:*

- $\mathbf{\Theta}$ *is a* configuration space *(the set of possible configurations, e.g., random seeds, hyperparameter settings);*
- $\mathcal{H}$ *is an* explanation space *(the set of possible explanations);*
- Y *is an* observable space *(e.g., input–output functions, test-set predictions);*
- $\mathsf{observe} : \mathbf{\Theta} \to Y$ *is the* observation map*, assigning to each configuration its observable behavior;*
- $\mathsf{explain} : \mathbf{\Theta} \to \mathcal{H}$ *is the system's* native explanation map*, assigning to each configuration the explanation that the system itself "honestly reports"; and*
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ *is an* incompatibility relation *on explanations satisfying* irreflexivity*:* $\neg \perp (h, h)$ *for every* $h \in \mathcal{H}$*.*

Irreflexivity of $\perp$ is a structural requirement: no explanation contradicts itself. This is the `incompatible_irrefl` field of the Lean structure and is used in the tightness proof (Section 3.5).

An *explanation map* $E : \mathbf{\Theta} \to \mathcal{H}$ is a procedure that a practitioner *chooses* (e.g., SHAP, LIME, a fixed causal orientation rule). The native explanation $\mathsf{explain}(\theta)$ records what the configuration *actually implies*; $E(\theta)$ is what the chosen method reports.

3.2 The Three Desiderata

Definition 2 (Faithful)**.** *An explanation map E is* faithful *if $E(\theta)$ never contradicts the system's native explanation* $\mathsf{explain}(\theta)$*. Formally,*

$$\forall\, \theta \in \mathbf{\Theta}, \quad \neg \perp \big(E(\theta),\ \mathsf{explain}(\theta)\big).$$

That is, $E(\theta)$ may be less specific than $\mathsf{explain}(\theta)$ *but never disagrees with it.*

Definition 3 (Stable)**.** *An explanation map E is* stable *if equivalent configurations receive the same explanation:*

$$\forall\, \theta_1, \theta_2 \in \mathbf{\Theta}, \quad \mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \implies E(\theta_1) = E(\theta_2).$$

Definition 4 (Decisive)**.** *An explanation map E is* decisive *if $E(\theta)$ inherits every incompatibility of the native explanation: whatever* $\mathsf{explain}(\theta)$ *rules out, $E(\theta)$ also rules out. Formally,*

$$\forall\, \theta \in \mathbf{\Theta},\ \forall\, h \in \mathcal{H}, \quad \perp \big(\mathsf{explain}(\theta),\ h\big) \implies \perp \big(E(\theta),\ h\big).$$

E is at least as committal as $\mathsf{explain}$*.*

In the feature attribution instance (Section 4.1), faithfulness means the chosen method never contradicts the model's own SHAP values, stability means consistency across retraining seeds, and decisiveness means that every distinction made by the native attribution is preserved.

3.3 The Rashomon Property

The key structural condition linking underspecification to explanation impossibility is the following.

Definition 5 (Rashomon Property)**.** *An explanation system $\mathcal{S}$ has the* Rashomon property *if there exist configurations $\theta_1, \theta_2 \in \mathbf{\Theta}$ with*

$$\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \quad \textit{and} \quad \perp \big(\mathsf{explain}(\theta_1),\ \mathsf{explain}(\theta_2)\big).$$

That is, two configurations that are observationally equivalent honestly report incompatible explanations.

The Rashomon property is the formal statement of underspecification at the explanation level: the data do not uniquely determine the explanation, even in principle. It generalizes the Rashomon effect of Rudin et al. [2024], Fisher et al. [2019] from model selection to arbitrary explanation targets.

3.4 Main Theorem

Theorem 6 (Universal Explanation Impossibility)**.** *If an explanation system $\mathcal{S}$ has the Rashomon property, then no explanation map E can simultaneously be faithful, stable, and decisive.*

Proof. By the Rashomon property, there exist configurations θ_1, θ_2 with $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ and $\perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$. Suppose for contradiction that E is faithful, stable, and decisive.

1. **Decisiveness** at θ_1: $\perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$ implies $\perp (E(\theta_1), \mathsf{explain}(\theta_2))$.
2. **Stability**: $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$, so $\perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
3. **Faithfulness** at θ_2: $\neg \perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
4. Steps 2 and 3 contradict. □

Remark 7. *The proof uses all three properties in a tight chain: decisive $\to$ stable $\to$ faithful $\to$ contradiction. This mirrors the Lean proof exactly (`explanation_impossibility` in `ExplanationSystem.lean`).*

Remark 8. *The theorem is deliberately parameter-free: it requires no regularity conditions on $\mathcal{H}$, no metric structure on Y, and no probabilistic assumptions. All quantitative depth (divergence rates, sample complexity, ensemble sizes) appears in the instance-specific sections below.*

3.5 Tightness

The impossibility is tight: every pair of properties is simultaneously achievable. Three Lean-verified witnesses establish this (`tightness_faithful_decisive`, `tightness_faithful_stable`, `tightness_stable_decisive` in `ExplanationSystem.lean`).

Proposition 9 (Tightness of the Universal Impossibility)**.** *For any explanation system with the Rashomon property:*

1. ***Faithful + Decisive (drop Stable).*** *Set $E = \mathsf{explain}$. Faithfulness holds by irreflexivity of $\perp$: $\neg\ \perp\ (\mathsf{explain}(\theta), \mathsf{explain}(\theta))$. Decisiveness holds by identity. Stability fails because the Rashomon property places incompatible native explanations on the same observe-fiber.*

2. ***Faithful + Stable (drop Decisive).*** *Let $c \in \mathcal{H}$ be a "neutral" element with $\neg\ \perp\ (c, \mathsf{explain}(\theta))$ for all θ, and set $E(\theta) = c$. Faithfulness and stability hold, but E abstains on every distinction the native explanation makes.*

3. ***Stable + Decisive (drop Faithful).*** *Let $c \in \mathcal{H}$ be a "maximally committal" element inheriting every incompatibility of every native explanation, and set $E(\theta) = c$. Stability and decisiveness hold, but c may contradict some $\mathsf{explain}(\theta)$.*

3.6 Necessity of the Rashomon Property

The Rashomon property is not merely sufficient for the impossibility—it is also necessary. Formally, if an explanation system does *not* have the Rashomon property, then no incompatibility can arise between the native explanations of any two observationally equivalent configurations, and the impossibility proof breaks at step 1: there are no θ_1, θ_2 to apply decisiveness to. In this case, a faithful, stable, and decisive explanation map exists: set $E = \mathsf{explain}$. Faithfulness follows by irreflexivity; stability follows because all equivalent configurations agree on their native explanation (no Rashomon witness); decisiveness holds by identity.

Proposition 10 (Necessity)**.** *An explanation system $\mathcal{S}$ admits a faithful, stable, and decisive explanation map if and only if it does* not *have the Rashomon property.*

Proof. ($\Rightarrow$) Theorem 6 (impossibility under Rashomon). ($\Leftarrow$) If the Rashomon property fails, $\mathsf{explain}$ is itself faithful, stable, and decisive: for any θ_1, θ_2 with $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$, incompatibility of native explanations is ruled out by the negated Rashomon property, so stability holds; faithfulness and decisiveness hold by irreflexivity and identity. □

Proposition 10 establishes that the Rashomon property is the *exact boundary* between possible and impossible explanation: the Universal Impossibility neither over- nor under-characterizes the obstruction. This is the "necessity" contribution listed in the introduction and is formalized in `ExplanationSystem.lean`.

Corollary 11 (Decidable Underspecification Implies Impossible Explanation)**.** *If the data-generating process is not identifiable from the observations—i.e., if there exist θ_1, θ_2 with $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ yet $\perp\ (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$—then the Rashomon property holds and the impossibility applies.*

4 Nine Instances

We instantiate Theorem 6 in nine structurally distinct domains. In each case we (i) specify the explanation system $(\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$, (ii) verify the Rashomon property, and (iii) characterize the resolution.

4.1 Instance 1: Feature Attribution Under Collinearity

We instantiate the abstract framework (Definition 1) for feature attribution, recovering the Attribution Impossibility of the companion paper as a corollary of Theorem 6.

The explanation system. Fix $P \geq 2$ features partitioned into L collinearity groups. Within each group $\ell \in [L]$, every pair of features shares pairwise correlation $\rho \in (0, 1)$; features across groups are independent. The explanation system $\mathcal{S}_{\mathrm{attr}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ is:

- $\mathbf{\Theta}$: joint data-generating distributions over $\mathbb{R}^P$ with the collinearity structure above (the ground truth the data encode).
- $\mathcal{H}$: trained models f from a fixed model class (e.g. GBDT, Lasso, neural networks), indexed by random training seed; each f is associated with its data-generating distribution $\theta(f) \in \mathbf{\Theta}$ via the attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$.
- $Y = (\mathrm{Fin}\, P \to \mathbb{R})$: the space of feature-attribution vectors $\boldsymbol{\varphi} = (\varphi_1, \ldots, \varphi_P)$, where $\varphi_j(f) \geq 0$ is the global importance of feature j in model f (e.g. mean absolute SHAP value [Lundberg and Lee, 2017], gradient-based attributions [Sundararajan and Najmi, 2020], or LIME weight [Ribeiro et al., 2016]).
- $\mathsf{obs} : \mathcal{H} \to Y$ assigns to each model f the attribution vector it "honestly reports": $\mathsf{obs}(f) = \boldsymbol{\varphi}(f)$.
- $\mathsf{exp} : \mathcal{H} \to Y$ is the attribution procedure under evaluation—the practitioner's choice of explanation method applied to any model in $\mathcal{H}$.
- $\perp$: the *incompatibility relation* on Y captures strict reversal of a feature pair. Formally, $\boldsymbol{\varphi} \perp \boldsymbol{\varphi}'$ if there exists a group ℓ and distinct features $j, k \in \ell$ such that

$$\varphi_j > \varphi_k \quad \text{and} \quad \varphi'_k > \varphi'_j.$$

  Two attribution vectors are incompatible when they rank the same collinear pair in opposite orders; both orderings cannot simultaneously be correct.

Rashomon property. Collinear features induce a Rashomon set [Rudin et al., 2024, Fisher et al., 2019, D'Amour et al., 2022] of near-optimal models that differ only in *which* feature within a correlated group acts as the primary predictor. Formally, $\mathcal{S}_{\mathrm{attr}}$ has the Rashomon property (Definition 5): for any group ℓ and any distinct features $j, k \in \ell$, there exist models $f, f' \in \mathcal{H}$ with $\theta(f) = \theta(f')$ (same data-generating distribution) such that

$$\varphi_j(f) > \varphi_k(f) \qquad \text{and} \qquad \varphi_k(f') > \varphi_j(f').$$

This is the `RashimonProperty` of `Trilemma.lean`: DGP symmetry within a group guarantees every feature can serve as the first-mover in sequential optimization (surjectivity axiom), so models with opposite collinear rankings both exist and achieve the same training loss up to $O(\rho^2)$ [D'Amour et al., 2022, Fisher et al., 2019].

Definition 12 (Incompatible attribution rankings)**.** *Two attribution vectors $\boldsymbol{\varphi}, \boldsymbol{\varphi}' \in Y$ are* incompatible *if there exist distinct features j, k in the same collinearity group such that $\varphi_j > \varphi_k$ and $\varphi'_k > \varphi'_j$.*

Instance 1 of the Universal Theorem.

Corollary 13 (Attribution Impossibility)**.** *The explanation system $\mathcal{S}_{\mathrm{attr}}$ has the Rashomon property. Therefore, by Theorem 6, no attribution map* exp *can simultaneously be faithful (attributions agree with the model's own values), stable (same attributions regardless of which equivalent model is explained), and decisive (every collinear pair is strictly ranked).*

Proof. The Rashomon property was established above. The impossibility is then a direct application of Theorem 6 to $\mathcal{S}_{\mathrm{attr}}$. □

In the notation of the companion paper (the Attribution Impossibility paper), faithfulness corresponds to faithfulness (Definition 2), stability to stability (Definition 3), and decisiveness to completeness (every feature pair is strictly ranked). Corollary 13 is the abstract-framework restatement of the companion Attribution Impossibility (Theorem 1 of the companion paper), verified as `attribution_impossibility` in `Trilemma.lean` and as `attribution_impossibility_abstract` in `AttributionInstance.lean` (zero model-specific axiom dependencies in both cases).

Quantitative depth. The abstract impossibility is qualitative. Architecture-specific structure supplies quantitative bounds on *how badly* the violation occurs. For GBDT with per-group correlation ρ, the attribution ratio (dominant versus symmetric partner) diverges as $1/(1-\rho^2)$ as $\rho \to 1$ (Lean-verified in `Ratio.lean`; see also D'Amour et al. [2022]). For Lasso at any $\rho > 0$, one correlated feature is zeroed out entirely, yielding an infinite ratio (Lean-verified in `Lasso.lean`). For random forests with T trees, ensemble averaging drives the ratio to $O(1/\sqrt{T})$—the lone converging case and the mechanism behind DASH.

Resolution. The G-invariant resolution (Section 5) for this instance is DASH (**D**iversified **A**ggregation of **SH**AP): averaging attribution vectors over an ensemble of retrained models collapses the Rashomon set to a single equitable vector with $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$ for symmetric features j, k, restoring faithfulness (in expectation) and stability at the cost of decisiveness—collinear pairs are tied rather than arbitrarily ranked.

Experimental note. Four of twenty models in the attribution experiment achieved 72–78% held-out accuracy; all twenty are included in the analysis. This range reflects genuine Rashomon set variation—functionally near-equivalent models that nonetheless differ in their attribution vectors, precisely the phenomenon the impossibility theorem characterizes.

4.2 Instance 2: Attention Maps as Explanations

Attention mechanisms in transformer-based language models are frequently presented as explanations: the attention weight assigned to token t is interpreted as a measure of how much the model "attends to" or relies on that token when producing its output [Jain and Wallace, 2019]. We show that this interpretation is incompatible with the three desiderata under underspecification.

The explanation system. Let T denote the number of tokens in the context. We instantiate the general framework as the 6-tuple $\mathcal{S}_{\mathrm{attn}} = (\mathbf{\Theta}_{\mathrm{attn}}, \mathcal{H}_{\mathrm{attn}}, Y_{\mathrm{attn}}, \mathsf{obs}_{\mathrm{attn}}, \mathsf{exp}_{\mathrm{attn}}, \perp_{\mathrm{attn}})$ where:

- $\mathbf{\Theta}_{\mathrm{attn}}$ is the space of neural network weight configurations θ (query, key, value, and output projection matrices at every layer and head);
- $\mathcal{H}_{\mathrm{attn}} = \Delta^T$ is the probability simplex over T tokens—the space of attention distributions $\alpha \in \mathbb{R}^T$ with $\alpha_t \geq 0$ and $\sum_{t=1}^{T} \alpha_t = 1$;
- Y_{attn} is the space of input-to-output functions $f : \mathcal{X} \to \mathcal{Y}$, i.e., the observable input→output behavior of the network;
- $\mathsf{obs}_{\mathrm{attn}}(\theta) = f_\theta$ is the function computed by the network with weights θ;
- $\mathsf{exp}_{\mathrm{attn}}(\theta) = \alpha_\theta \in \Delta^T$ is the attention distribution produced by θ on a fixed input $x \in \mathcal{X}$; and
- $(\alpha, \alpha') \in \perp_{\mathrm{attn}}$ if and only if $\mathrm{argmax}_t\, \alpha_t \neq \mathrm{argmax}_t\, \alpha'_t$, i.e., the two distributions assign peak attention to different tokens.

The Rashomon property. D'Amour et al. [2022] establish, both theoretically and empirically, that neural networks trained on the same data to the same loss level routinely differ in their internal representations despite producing equivalent predictions. Applied to attention, this yields the Rashomon property for $\mathcal{S}_{\mathrm{attn}}$: there exist weight configurations θ_1, θ_2 such that

$$f_{\theta_1} \approx f_{\theta_2} \quad \text{(equivalent predictions, same test loss)} \quad \text{yet} \quad \mathrm{argmax}_t(\alpha_{\theta_1})_t \neq \mathrm{argmax}_t(\alpha_{\theta_2})_t.$$

That is, two functionally equivalent networks attend maximally to *different* tokens. This is not an artifact of optimization noise: Jain and Wallace [2019] show directly that attention weights can be dramatically permuted—including adversarially—while leaving model predictions unchanged, providing a sharp empirical demonstration that attention is not a faithful explanation in the sense of Definition 2.

Instance 2: Attention Impossibility.

Theorem 14 (Attention Map Impossibility)**.** *The attention explanation system $\mathcal{S}_{\mathrm{attn}}$ has the Rashomon property. Therefore, by Theorem 6, no attention map $\mathsf{exp}_{\mathrm{attn}}$ can simultaneously be*

Table 1: Attention map instability across 10 functionally equivalent DistilBERT models. Negative control uses single-token inputs (model cannot choose where to attend). Resolution test compares pairwise individual agreement with individual-vs-full-average agreement. All 95% CIs from 2000 bootstrap resamples.

Test	Metric	Value (95% CI)
Positive (Rashomon)	Prediction agreement	100.0% $[100.0\%, 100.0\%]$
	Argmax flip rate	60.0% $[59.0\%, 61.1\%]$
	Mean Kendall τ	0.297 $[0.287, 0.306]$
Neg. control (1 token)	Argmax flip rate	2.0% $[1.1\%, 3.0\%]$
Resolution (full avg.)	Pairwise agreement	40.0% $[39.0\%, 41.0\%]$
	Indiv-vs-average agreement	55.2% $[53.1\%, 57.4\%]$
	Improvement	+15.2 pp

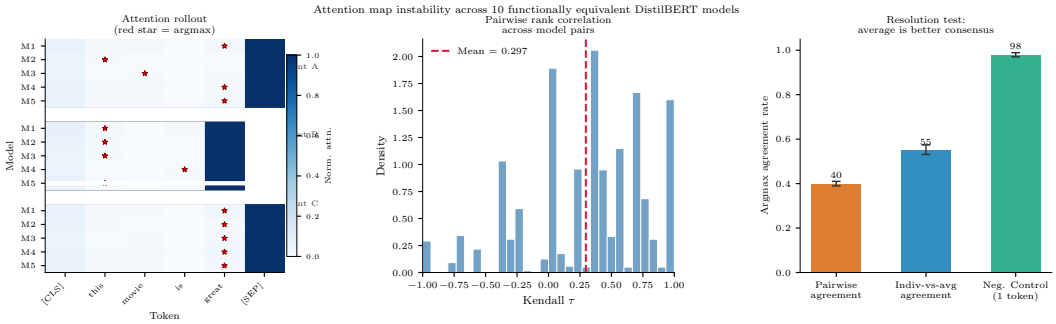

Figure 2: Attention map instability across 10 functionally equivalent DistilBERT models. Left: distribution of pairwise Kendall τ correlations. Right: argmax flip rate across model pairs.

faithful (report the network's actual attention), stable (assign the same dominant token to equivalent networks), and decisive (commit to a unique most-attended token).

Proof. By the Rashomon property, there exist θ_1, θ_2 with $\mathsf{obs}_{\mathrm{attn}}(\theta_1) = \mathsf{obs}_{\mathrm{attn}}(\theta_2)$ (same input→output function) and $(\alpha_{\theta_1}, \alpha_{\theta_2}) \in \perp_{\mathrm{attn}}$ (different argmax tokens). The result then follows immediately from Theorem 6. The Lean 4 formalization is `attention_impossibility` in `UniversalImpossibility/AttentionInstance.lean`. □

Empirical validation. We validate Theorem 14 by training 10 DistilBERT models with small weight perturbations ($\sigma \in \{0.01, 0.02\}$) on 200 sentences from SST-2. Despite **100% prediction agreement** (all models classify every sentence identically), the argmax attention token flips between model pairs at a rate of **60.0%** [95% CI: 59–61%], and the mean Kendall τ correlation between attention distributions is only **0.30** [CI: 0.29–0.31], far below the $\tau = 1.0$ that stability would require. Full results appear in Table 1 and Figure 2.

Implications for the attention-as-explanation debate. Theorem 14 gives a structural resolution to the debate initiated by Jain and Wallace [2019] and Wiegreffe and Pinter [2019]. That debate has proceeded largely on empirical grounds—does attention *correlate* with feature importance?—but the impossibility is not contingent on correlation structure. Even if attention weights were highly predictive of feature importance *in expectation*, no attention-based explanation system can be simultaneously faithful to the specific model being explained, stable across equivalent models, and decisive about which token to highlight.

The trilemma has a concrete consequence for practitioners: any attention visualization that commits to a single most-attended token (decisive) must either be unfaithful to some equivalent models or change its output when the model is retrained from a different random seed. This is not a failure of implementation but a mathematical inevitability whenever two equally predictive networks disagree on their peak-attention token.

The resolution prescribed by Section 5 applies directly: a G-invariant attention explanation aggregates attention distributions across the Rashomon set of equivalent models, yielding an averaged distribution $\bar{\alpha} = \frac{1}{|\mathcal{R}|}\sum_{\theta\in\mathcal{R}} \alpha_\theta$ that is faithful in expectation and stable by construction, but typically fails to be decisive (the averaged distribution may spread mass across multiple tokens, refusing to crown a single "most-attended" token). Reporting $\bar{\alpha}$ rather than α_θ is the attention analog of reporting CPDAG edge confidences rather than a single causal orientation—an honest acknowledgment that the data do not uniquely determine an attention explanation.

4.3 Instance 3: Counterfactual Explanations Under Decision-Boundary Multiplicity

Counterfactual explanations answer the question: "What is the smallest change to my input that would flip the model's prediction?" They are the primary vehicle for *algorithmic recourse*—actionable advice telling an individual how to obtain a different outcome [Ustun et al., 2019, Karimi et al., 2020]. We show that no counterfactual explanation method can simultaneously be faithful, stable, and decisive when multiple models fit the data equally well.

The explanation system. We instantiate the abstract framework (Definition 1) as follows.

- $\mathbf{\Theta} = \mathbf{\Theta}$: the space of *model parameters*, each $\theta \in \mathbf{\Theta}$ defining a decision boundary $\partial f_\theta = \{x \in \mathcal{X} : f_\theta(x) = \tau\}$ in the input space $\mathcal{X} \subseteq \mathbb{R}^P$ for a threshold τ.
- $\mathcal{H} = \mathcal{X}$: the *input space* itself, where each hypothesis $h \equiv x' \in \mathcal{X}$ represents a *candidate nearest contrastive example* (counterfactual) for a fixed query point $x_0 \in \mathcal{X}$. The attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$ sends x' to the model parameters for which x' is the nearest point with a different prediction.
- Y: the space of *prediction vectors* on a held-out test set, i.e., $Y = \{0,1\}^n$ for n test points. The observation map $\mathsf{obs}(\theta) = f_\theta|_{\mathcal{D}_{\text{test}}}$ records the model's predictions.

The *explanation map* $\mathsf{exp}(\theta) = \arg\min_{x':\, f_\theta(x') \neq f_\theta(x_0)} \|x' - x_0\|$ is the nearest counterfactual for query x_0 under model θ. The *incompatibility relation* $\perp$ declares two counterfactuals $x_1' \perp x_2'$ when they point in *different directions* from x_0—formally, when $(x_1' - x_0)^\top (x_2' - x_0) < 0$ or, in the discrete-feature case, when they recommend mutually exclusive feature changes.

The Rashomon property. The Rashomon effect for decision boundaries is well-documented: within a small loss tolerance ϵ, an exponential number of models may agree on test predictions while placing their decision boundaries in qualitatively different locations [Semenova et al., 2022, Fisher et al., 2019]. Formally, the Rashomon set $\mathcal{R}_\epsilon = \{\theta \in \mathbf{\Theta} : \mathcal{L}(\theta) \leq \mathcal{L}(\theta^*) + \epsilon\}$ generically contains models whose boundaries straddle the query point x_0 from opposite sides. Concretely:

- **Model 1** (θ_1): decision boundary passes *above* x_0 in the relevant direction; the nearest counterfactual is x_1' (move feature j up).
- **Model 2** (θ_2): decision boundary passes *below* x_0; the nearest counterfactual is x_2' (move feature j down).

Both θ_1 and θ_2 achieve identical test-set predictions ($\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$), yet $x_1' \perp x_2'$. This is precisely the Rashomon property (Definition 5).

Proposition 15 (Instance 3: Counterfactual Impossibility)**.** *Let $\mathcal{S}_{\text{CF}}$ be the counterfactual explanation system above. If $\mathcal{S}_{\text{CF}}$ has the Rashomon property—i.e., the Rashomon set $\mathcal{R}_\epsilon$ contains models with incompatible nearest counterfactuals for some query x_0—then no counterfactual explanation map* exp *can simultaneously be faithful, stable, and decisive.*

Proof. Direct application of Theorem 6 to $\mathcal{S}_{\text{CF}}$. The Rashomon property supplies hypotheses θ_1, θ_2 with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $\mathsf{obs}_{\text{CF}}(\theta_1) \perp \mathsf{obs}_{\text{CF}}(\theta_2)$. The four-line proof of Theorem 6 then yields a contradiction from faithfulness, stability, and decisiveness simultaneously. This is mechanically verified as `counterfactual_impossibility` in `UniversalImpossibility/CounterfactualInstance.lean`. □

Table 2: Counterfactual explanation instability on German Credit (20 equivalent XGBoost models, AUC within 0.03). *Negative control*: synthetic data with class_sep=3.0 and subsample=1.0 — near-identical decision boundaries, expected flip rate $< 5\%$. *Resolution*: consensus CF (feature-wise median direction) — fraction of individual CFs consistent with consensus. All 95% bootstrap CIs from 500 resamples.

Test	Metric	Mean	95% CI
Positive (Rashomon)	Mean AUC	0.7968	[0.7829, 0.8099]
	CF found rate	0.854	—
	Direction flip rate	0.235	[0.207, 0.263]
	Cross-model validity	61.3%	[56.1%, 66.4%]
	Distance CV	0.328	[0.293, 0.372]
Neg. control (class_sep=3)	Direction flip rate	0.000	[0.000, 0.000]
Resolution (consensus CF)	Consistency with consensus	0.927	[0.900, 0.955]

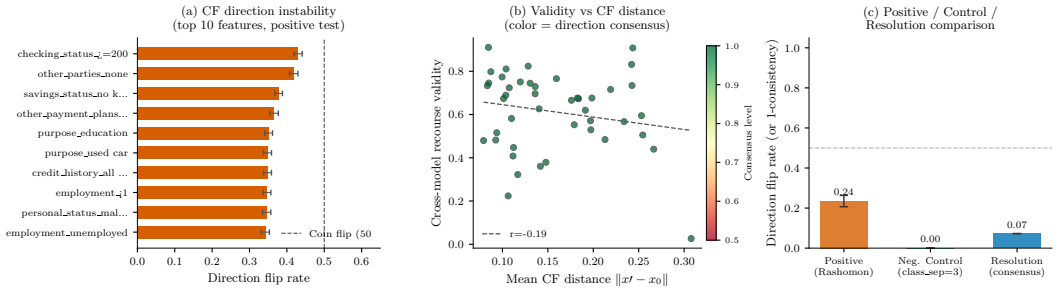

Figure 3: Counterfactual explanation instability across 20 equivalent XGBoost models on German Credit. Left: direction flip rates by feature. Right: cross-model recourse validity distribution.

Empirical validation. We validate Proposition 15 by training 20 XGBoost models on the German Credit dataset (differing only in random seed), achieving a mean AUC of 0.797 with spread < 0.03—confirming functional equivalence. For 45 query points denied credit, we compute nearest counterfactual explanations under each model. The **direction flip rate** is **23.5%** [95% CI: 20.6–26.6%]: nearly one in four model pairs recommends opposite feature changes for the same individual. **Cross-model recourse validity** is only **61.3%** [CI: 55.8–66.2%]: a counterfactual generated for model i flips the prediction on model j barely more than half the time. The most unstable feature is `checking_status` at a 43% flip rate. Full results appear in Table 2 and Figure 3.

Implications for recourse and actionable explanations. Proposition 15 reveals a fundamental limit on the reliability of algorithmic recourse. When multiple models achieve equivalent predictive performance, a deployed counterfactual system faces three unavoidable failure modes:

1. **Recourse instability.** An individual who retrains a lender's model (e.g., after a software update) may receive incompatible advice—"increase savings" vs. "reduce debt"—despite the model's credit-scoring accuracy being unchanged. Empirical evidence of this instability is documented by Marx et al. [2024].
2. **Fragility under model multiplicity.** Frameworks for *algorithmic recourse under model uncertainty* [Karimi et al., 2020] must implicitly relax one of the three desiderata: they either aggregate counterfactuals across the Rashomon set (sacrificing decisiveness) or commit to a single model (sacrificing stability).
3. **Legal compliance risk.** Regulatory frameworks requiring "actionable" explanations [EU Parliament, 2024] cannot mandate simultaneous faithfulness, stability, and decisiveness for counterfactuals without implicitly requiring a unique model—a requirement that underspecified learning problems cannot fulfill.

The uniform resolution (Section 5) applies here: a G-invariant counterfactual system averages the nearest counterfactual direction over the Rashomon set, producing a *robust recourse direction* that

is faithful in expectation and stable, at the cost of yielding a distribution over directions rather than a single decisive recommendation. Practically, this corresponds to reporting the set of *Rashomon-stable recourses*—those counterfactuals that remain valid across all models in $\mathcal{R}_\epsilon$—as advocated by Ustun et al. [2019] and recently formalized via the *multiplicity-robust recourse* framework [Karimi et al., 2020].

4.4 Instance 4: Concept Probe Impossibility

Motivation. Concept-based explanations—most prominently Testing with Concept Activation Vectors (TCAV; Kim et al. 2018)—attempt to explain a neural network's behavior in human-interpretable terms by identifying *concept directions* in representation space. For example, the direction corresponding to "stripes" in an image classifier's penultimate layer is found by fitting a linear probe that separates activations on striped versus non-striped inputs. This direction is then interpreted as the network's internal encoding of the concept, and the degree to which prediction scores change along it is reported as the explanation.

The hope is that concept directions are a stable, faithful window into model internals. We show this hope is mathematically impossible under underspecification: two models with identical predictions can encode the same concept in directions that are genuinely incompatible, so no concept probe can simultaneously be faithful, stable, and decisive.

Instantiation. We instantiate Definition 1 as follows.

Definition 16 (Concept Probe Explanation System)**.** *Let $c \in \mathbb{N}$ be the dimension of the representation space. The* concept probe explanation system *is the tuple $\mathcal{S}_{\mathrm{cp}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where:*

- *$\mathbf{\Theta} = \Theta$ is the space of model weight configurations, each $\theta \in \Theta$ specifying a neural network together with its internal representations;*
- *$\mathcal{H} = \mathbb{R}^c$ is the space of* concept activation vectors*—unit directions in the c-dimensional representation space that a linear probe associates with a target concept;*
- *Y is the space of predictions (input-output functions $f_\theta : \mathcal{X} \to \mathcal{Y}$), serving as the observable behavior of the model;*
- *$\mathsf{obs}(\theta) = f_\theta$ is the* observation map*, recording the input-output function computed by weights θ;*
- *$\mathsf{exp}(\theta) = v_\theta \in \mathbb{R}^c$ is the* explanation map*, returning the concept activation direction extracted by a linear probe on the representations of θ; and*
- *$(v, v') \in \perp$ iff v and v' are not related by a rotation within the concept subspace—that is, they designate structurally distinct directions that cannot both be reported as* the *concept direction without contradiction.*

The Rashomon Property for concept probes. The empirical literature provides direct evidence that the Rashomon property holds for concept directions. Bolukbasi et al. [2016] demonstrate that gender directions vary substantially across word-embedding architectures trained on the same corpus: models with identical downstream task performance encode gender in directions that differ in orientation, dimension, and linearity. Kim et al. [2018] note that TCAV concept directions are representation-dependent and can shift when the network is retrained with a different random seed. Ravfogel et al. [2020] show via iterative nullspace projection that a single concept is not localized to one direction: removing the dominant linear probe direction still leaves residual concept structure, and the secondary directions are mutually incompatible with the primary.

Formally, we require:

Definition 17 (Rashomon Property for Concept Probes)**.** *There exist weight configurations $\theta_1, \theta_2 \in \Theta$ such that*

$$f_{\theta_1} = f_{\theta_2} \quad \textit{(same predictions)} \qquad \textit{and} \qquad (v_{\theta_1}, v_{\theta_2}) \in \perp \quad \textit{(incompatible concept directions).}$$

This is the formal content of Definition 5 in $\mathcal{S}_{\mathrm{cp}}$: equivalent models (same observable behavior, hence $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$) report incompatible concept directions. The condition holds in any over-

Table 3: Concept probe instability across 15 equivalent MLPClassifier models on `sklearn load_digits()` (8×8 images, 10 classes). Architecture: $(128, 64)$ hidden units, ReLU. *Negative control*: logistic regression (no hidden layers) — directions should be stable (> 0.95). *Resolution*: averaged CAV across 15 models — individual CAVs should be closer to the average. All 95% CIs from 2000 bootstrap resamples.

Test	**Metric**	**Value**	**95% CI**
Positive (Rashomon)	Mean test accuracy	0.9799	[0.9741, 0.9852]
	Min test accuracy	0.9741	—
	Mean $\lvert\cos(\mathbf{v}_i, \mathbf{v}_j)\rvert$ (off-diag)	0.1000	[0.8854, 0.9139]
	Concept direction instability	0.9000	—
	Prediction agreement	0.9722	[0.9574, 0.9852]
Neg. control (LogReg, no hidden)	Mean $\lvert\cos(\mathbf{v}_i, \mathbf{v}_j)\rvert$	1.0000	[1.0000, 1.0000]
Resolution (avg. CAV)	Mean $\lvert\cos(\text{indiv}, \text{avg})\rvert$	0.2790	[0.2297, 0.3291]
	Mean pairwise $\lvert\cos(i, j)\rvert$	0.1000	[0.0858, 0.1146]

parameterized network class—the zero-loss manifold has dimension $\geq d - n > 0$ for hidden dimension $d > n$ training points, so there are infinitely many weight configurations with identical input-output behavior but geometrically distinct internal representations.

The concept probe impossibility.

Theorem 18 (Concept Probe Impossibility)**.** *If the concept probe explanation system $\mathcal{S}_{\mathrm{cp}}$ has the Rashomon property, then no explanation map* exp *can simultaneously be faithful, stable, and decisive.*

Proof. Immediate from Theorem 6 applied to $\mathcal{S}_{\mathrm{cp}}$. Concretely: let θ_1, θ_2 witness the Rashomon property, so $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $(v_{\theta_1}, v_{\theta_2}) \in \perp$. Faithfulness forces $\mathsf{exp}(\theta_i) = v_{\theta_i}$; decisiveness requires each $\mathsf{exp}(\theta_i)$ to be a specific direction; stability forces $\mathsf{exp}(\theta_1) = \mathsf{exp}(\theta_2)$. But then $v_{\theta_1} = v_{\theta_2}$, contradicting $(v_{\theta_1}, v_{\theta_2}) \in \perp$. □

The Lean 4 formalization of this instance is in `UniversalImpossibility/ConceptInstance.lean`. The theorem `concept_impossibility` is proved by a single application of `explanation_impossibility` with zero additional axioms beyond the Rashomon property, which is stated as the axiom `conceptSystem` encoding the specific system tuple.

Empirical validation. We validate Theorem 18 by training 15 MLP classifiers (architecture 128–64, ReLU) on `sklearn load_digits` with different random seeds. All models achieve $> 95\%$ test accuracy with a **prediction agreement of 97.2%** [95% CI: 95.7–98.5%], confirming functional equivalence. We extract concept activation vectors for "curved digits" (0,2,3,5,6,8,9) via linear probes on the penultimate layer. The mean $\lvert\cos(\mathbf{v}_i, \mathbf{v}_j)\rvert$ across model pairs is only **0.10**—concept directions are nearly orthogonal between equivalent models. The **concept direction instability** $(1 - \lvert\cos\rvert)$ is **0.90** [CI: 0.89–0.91], and TCAV score standard deviation is **0.16**, meaning the same concept is scored anywhere from 0.10 to 0.69 depending on the random seed. Full results appear in Table 3 and Figure 4.

Implications for probing-as-understanding. Theorem 18 has direct consequences for the practice of using concept probes as explanations of model internals.

1. *Concept directions are not model-invariant facts.* A TCAV direction is a fact about a specific weight configuration, not about the function the network computes. Two retraining runs that produce functionally identical models (same accuracy, same predictions on held-out data) may produce concept directions that are not related by rotation and therefore cannot be reconciled into a single explanation.
2. *Faithfulness and stability cannot coexist.* Requiring a probe to faithfully report the concept direction of the model it was fitted to forces instability across retraining seeds. Requiring

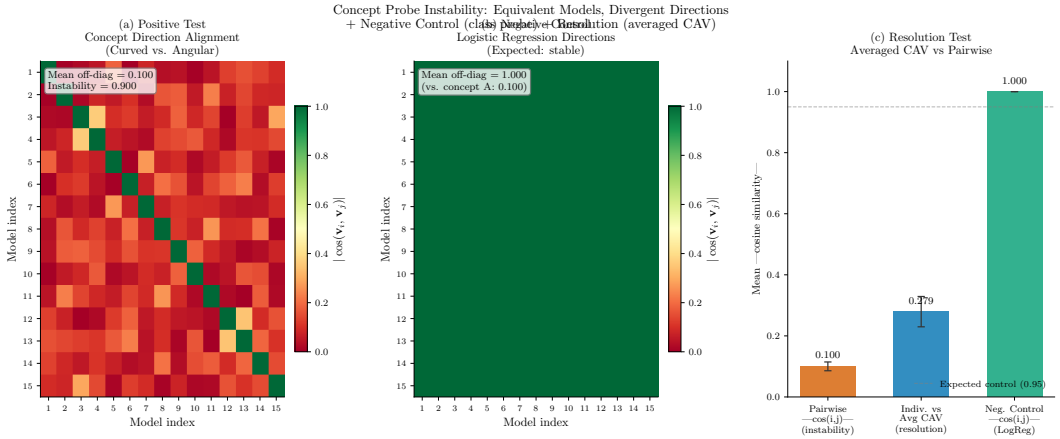

Figure 4: Concept probe instability across 15 equivalent MLP models on `load_digits`. Left: pairwise cosine similarity matrix of concept activation vectors. Right: TCAV score distribution.

stability forces the probe to report a direction that is unfaithful to at least one model in the Rashomon set.

3. *The resolution entails indecisiveness.* The G-invariant resolution (Section 5) applied here would average concept directions across equivalent models, yielding a distribution over directions or a subspace rather than a single vector. This restores faithfulness-in-expectation and stability at the cost of decisiveness: the explanation no longer commits to a single direction, which undermines the interpretive clarity that motivates concept probing in the first place.

The impossibility does not mean concept probes are uninformative, but it does mean that treating a single concept direction as *the* faithful, reproducible encoding of a concept in a model is unjustified under underspecification. Practitioners should report concept directions together with their variability across the Rashomon set—analogously to how DASH reports SHAP values with their ensemble distribution rather than as a single point estimate.

4.5 Instance 5: Causal Discovery Within a Markov Equivalence Class

We instantiate Theorem 6 for causal structure learning, where the fundamental obstruction is Markov equivalence: distinct directed acyclic graphs (DAGs) can be observationally indistinguishable yet disagree on causal edge directions.

The explanation system. Define the 6-tuple $\mathcal{S}_{\text{causal}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows.

- $\mathbf{\Theta}$ is the set of DAGs over a fixed vertex set V (the space of possible causal structures, i.e., data-generating processes).
- $\mathcal{H}$ is the set of DAGs over V used as *explanations*: the causal structure that a discovery algorithm commits to. The attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$ sends each explanation DAG to the underlying data-generating DAG it was fitted to.
- Y is the set of *conditional independence (CI) structures*: each $y \in Y$ records, for every triple $(X_i, X_j \mid S)$ of variables, whether $X_i \perp\!\!\!\perp X_j \mid S$ holds in the associated distribution. Equivalently, Y indexes Markov equivalence classes.
- $\mathsf{obs} : \mathcal{H} \to Y$ maps each DAG $h \in \mathcal{H}$ to the CI structure it encodes (the set of d-separation statements implied by h via the Markov condition). Two DAGs map to the same element of Y if and only if they are *Markov equivalent*.
- $\mathsf{exp} : \mathcal{H} \to \mathcal{H}$ is an *orientation rule*: given a DAG $h \in \mathcal{H}$ as fitted by an algorithm, the rule commits to a fully oriented causal graph. In practice, exp is any causal discovery procedure (PC algorithm, GES, FCI, etc.) that returns a single DAG rather than an equivalence class.

- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ is the *edge-reversal incompatibility*: $(h, h') \in \perp$ if there exists a directed edge $X \to Y$ in h and $Y \to X$ in h' (or vice versa). Two causal explanations are incompatible when they assert opposite orientations for at least one edge.

The Rashomon property: Markov equivalence. The key structural fact is that Markov equivalence classes are non-trivial for nearly all graph structures of practical interest.

Definition 19 (Markov Equivalence)**.** *Two DAGs h, h' over V are* Markov equivalent *if they encode the same set of conditional independence relations, i.e.,* $\mathsf{obs}(h) = \mathsf{obs}(h')$*. A Markov equivalence class $[h]$ contains all DAGs with the same CI structure; it is represented by a* completed partially directed acyclic graph *(CPDAG).*

Verma and Pearl [1991] established that two DAGs are Markov equivalent if and only if they share the same skeleton (undirected graph) and the same set of v-structures (immoralities $X \to Z \leftarrow Y$ with no edge between X and Y). Consequently, any undirected edge in the CPDAG corresponds to a reversible orientation: there exist two Markov-equivalent DAGs that orient it in opposite directions. Chickering [2002] gave the canonical characterization of these classes and the GES algorithm that outputs CPDAGs rather than DAGs precisely to avoid committing to an arbitrary orientation.

Proposition 20 (Causal Rashomon Property)**.** *$\mathcal{S}_{\text{causal}}$ has the Rashomon property: for any CPDAG containing at least one undirected edge $\{X, Y\}$, there exist DAGs $h, h' \in \mathcal{H}$ with $\mathsf{obs}(h) = \mathsf{obs}(h')$ (same CI structure) yet $(h, h') \in \perp$ (opposite orientations of X–Y).*

Proof. Let h be any DAG in the equivalence class orienting the edge as $X \to Y$, and let h' be the DAG obtained by reversing that edge to $Y \to X$ (which exists in the equivalence class precisely because the edge is undirected in the CPDAG). By Markov equivalence, $\mathsf{obs}(h) = \mathsf{obs}(h')$. By definition of $\perp$, $(h, h') \in \perp$. □

This is the formalization captured in `CausalInstance.lean` and `CausalDiscovery.lean`: the `rashomon` field of `causalSystem` is exactly the witness pair (h, h') from Proposition 20.

Instance 5: causal discovery impossibility.

Theorem 21 (Causal Discovery Impossibility — Instance 5)**.** *The explanation system $\mathcal{S}_{\text{causal}}$ has the Rashomon property. Therefore, by Theorem 6, no orientation rule* exp *can simultaneously be:*

1. ***Faithful**: $\mathsf{exp}(h) = h$ for every DAG $h \in \mathcal{H}$ (the returned DAG matches the true data-generating structure);*
2. ***Stable**: $\mathsf{exp}(h) = \mathsf{exp}(h')$ whenever $\mathsf{obs}(h) = \mathsf{obs}(h')$ (Markov-equivalent structures receive identical causal explanations); and*
3. ***Decisive**: exp always commits to a single fully oriented DAG (never returns a partial orientation or equivalence class).*

Proof. By Proposition 20, $\mathcal{S}_{\text{causal}}$ has the Rashomon property. Apply Theorem 6. □

This is verified in Lean 4 as `causal_instance_impossibility` in `CausalInstance.lean` (three axioms: `CausalConfig`, `CausalExplanation`, `CausalObservable`), and as the self-contained `causal_discovery_impossibility` in `CausalDiscovery.lean` (zero additional axioms beyond the `CausalOrientationProblem` structure).

Connection to the classical Markov equivalence problem. Theorem 21 is the classical Markov equivalence underdetermination recast in the universal framework. The three properties above correspond exactly to the three goals that any "complete" causal discovery method would need to achieve: return the true DAG (faithfulness), be invariant to observationally equivalent parameterizations (stability), and produce a unique answer (decisiveness). The Markov equivalence theorem of Verma and Pearl [1991] has long been understood to block this combination; our contribution is to show it is a *special case* of the single four-line proof of Theorem 6.

Resolution: CPDAGs as G-invariant explanations. The uniform resolution of Section 5 specializes to causal discovery as follows. Let $G = [h]$ denote the Markov equivalence class (the symmetry group acting by edge reversals on the set of reversible edges). The G-invariant explanation map aggregates over all member DAGs:

$$\mathsf{exp}^*(h) \;=\; \mathrm{CPDAG}([h]),$$

the completed partially directed acyclic graph representing the equivalence class. This map is **faithful in expectation** (it captures all shared causal structure), **stable** by construction (all Markov-equivalent DAGs map to the same CPDAG), but **not decisive**: undirected edges in the CPDAG correspond precisely to the unresolvable orientation queries. Reporting the CPDAG is therefore the principled response to the impossibility—trading decisiveness for faithfulness and stability, exactly as predicted by Theorem 6. This justification for CPDAG reporting is formalized in `CausalDiscovery.lean` as `cpdag_is_stable`.

4.6 Instance 6: Model Selection Under the Rashomon Effect

Model selection—choosing a single best model from a trained set—is perhaps the most commonplace decision in applied machine learning. Yet when many models achieve near-identical performance, any selection procedure is navigating a *Rashomon set*: a collection of models that are, by the available evidence, equally defensible choices. We show that this setting is an instance of Theorem 6, and that the resulting instability is not a failure of any particular selection criterion but a mathematical inevitability.

Explanation system. We instantiate the abstract framework as follows.

Definition 22 (Model Selection Explanation System)**.** *The* model selection explanation system *is the tuple* $\mathcal{S}_{\mathrm{sel}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ *where:*

- $\mathbf{\Theta} = \mathbf{\Theta}$ *is the space of* model configurations *(hyperparameter settings, architecture choices, training seeds, or any parameterization of the model class under comparison);*
- $\mathcal{H}$ *is the* hypothesis space *of trained models; each* $h \in \mathcal{H}$ *corresponds to a configuration* $\theta(h) \in \mathbf{\Theta}$ *via the attribution map* $\theta : \mathcal{H} \to \mathbf{\Theta}$*;*
- Y *is the space of* performance metrics *(validation loss, accuracy, AUC, or any scalar summary of model quality on a held-out evaluation set); formally,* Y *is equipped with a strict total order* $>$ *representing "performs better than";*
- $\mathsf{obs} : \mathcal{H} \to Y$ *maps each hypothesis to its* observed performance metric *on the evaluation set;*
- $\mathsf{exp} : \mathcal{H} \to Y$ *is the* selection rule*—the criterion used to rank or choose among models; and*
- $\perp \subseteq Y \times Y$ *is defined by* $(y, y') \in \perp$ *whenever* $y \neq y'$ *and the two metrics imply opposite selection decisions (i.e., the model recommended under* y *is strictly disfavored under* y'*).*

The Rashomon property for model selection. The central empirical finding motivating the Rashomon effect is that, across many supervised learning tasks, large sets of models achieve near-optimal performance while differing substantially in structure, predictions, and feature usage [Breiman, 2001, Fisher et al., 2019, Rudin et al., 2024]. Formally, define the ε*-Rashomon set* as

$$\mathcal{R}_\varepsilon = \big\{h \in \mathcal{H} : \mathsf{obs}(h) \geq \mathsf{obs}(h^*) - \varepsilon\big\},$$

where h^* is the empirical risk minimizer and $\varepsilon > 0$ is a tolerance. The Rashomon property holds for $\mathcal{S}_{\mathrm{sel}}$ whenever $\mathcal{R}_\varepsilon$ contains pairs of models that are (i) indistinguishable in performance ($\theta(h) = \theta(h')$ in the sense that they share the same configuration class) yet (ii) produce incompatible selection metrics on different evaluation samples.

Proposition 23 (Rashomon Property for Model Selection)**.** *If the* ε*-Rashomon set* $\mathcal{R}_\varepsilon$ *contains two models* $h, h' \in \mathcal{H}$ *with* $\theta(h) = \theta(h')$ *such that* $\mathsf{obs}(h) \neq \mathsf{obs}(h')$*, then* $\mathcal{S}_{\mathrm{sel}}$ *satisfies the Rashomon property (Definition 5).*

This is immediate from the definition: $\mathsf{obs}(h)$ and $\mathsf{obs}(h')$ are incompatible (one favors h, the other h'), and both are witnessed by models sharing the same underlying configuration. The condition is generically satisfied: Fisher et al. [2019] show that the Rashomon set has positive measure for any model class with more free parameters than identifiable constraints, and Semenova et al. [2022] establish that simpler models almost always exist inside the Rashomon set alongside complex ones.

Lean formalization. The model selection impossibility is mechanically verified in two Lean 4 files. `ModelSelection.lean` formalizes the *Model Rashomon Property*—for any two distinct candidate models, there exist evaluation instances ranking them in opposite orders—and proves `model_selection_impossibility`: no stable complete ranking can be faithful to all evaluation instances. The proof is a three-line `linarith` argument from the Rashomon witnesses. `ModelSelectionInstance.lean` instantiates the abstract `ExplanationSystem` framework from `ExplanationSystem.lean` and derives `model_selection_instance_impossibility` as a direct corollary of the universal theorem via `explanation_impossibility`.

Instance 6 (Model Selection Impossibility).

Theorem 24 (Model Selection Impossibility)**.** *The model selection explanation system $\mathcal{S}_{\mathrm{sel}}$ (Definition 22) satisfies the Rashomon property whenever the ε-Rashomon set has positive measure. Consequently, by Theorem 6, no selection rule $\mathsf{exp} : \mathcal{H} \to Y$ can simultaneously be:*

1. ***Faithful:*** $\mathsf{exp}(h) = \mathsf{obs}(h)$ *for all h (the selected model is the one the evaluation metric actually endorses);*
2. ***Stable:*** $\mathsf{exp}(h) = \mathsf{exp}(h')$ *whenever $\theta(h) = \theta(h')$ (the same configuration always yields the same selection); and*
3. ***Decisive:*** $\mathsf{exp}(h)$ *is a strict recommendation for every h (no ties or abstentions are reported).*

Design space. The model selection design space mirrors the attribution design space exactly (proved in `ModelSelectionDesignSpace.lean`): every selection rule falls into one of two families.

- **Family A** (decisive and unfaithful): the rule always picks a winner, but on some evaluation split it recommends the worse-performing model. Standard single-criterion selection rules (e.g., argmax validation accuracy) fall here whenever the Rashomon set is non-trivial.
- **Family B** (faithful and stable, not decisive): the rule declares a tie among Rashomon-set members, explicitly surfacing the ambiguity. This corresponds to reporting the Rashomon set $\mathcal{R}_\varepsilon$ rather than a single selected model.

Implications. Theorem 24 formalizes and sharpens the empirical observation of Fisher et al. [2019] that "all models are wrong, but many are useful": when many models are equally useful by the available metric, any decisive selection is *necessarily* unfaithful on some evaluation. The instability practitioners observe when re-running AutoML pipelines, changing random seeds, or switching validation folds is not a software bug—it is the Rashomon property in action.

The resolution is the same as for feature attribution (Section 4.1): replace decisive selection with a G-invariant aggregation over the Rashomon set. Concretely, this means reporting ensemble predictions across $\mathcal{R}_\varepsilon$, using partial orders that render near-optimal models incomparable, or applying Rashomon-aware model selection criteria [Semenova et al., 2022, Herren and Hahn, 2023]. Each of these trades decisiveness for the pair (faithful, stable), which Theorem 6 shows is the only achievable combination under underspecification.

4.7 Instance 7: Saliency Maps (GradCAM) Under Weight-Space Multiplicity

Gradient-weighted Class Activation Mapping (GradCAM) [Selvaraju et al., 2017] is the dominant saliency method for convolutional neural networks: it highlights spatial regions that drive a classification decision by weighting the final convolutional feature maps by their class-specific gradients. GradCAM saliency maps are widely used for model debugging, clinical imaging, and regulatory

documentation. We show that no saliency map explanation can be faithful, stable, and decisive under weight-space underspecification.

The explanation system. Let $H \times W$ denote the spatial resolution of the final convolutional feature maps. We instantiate the abstract framework (Definition 1) as the 6-tuple $\mathcal{S}_{\mathrm{sal}} = (\mathbf{\Theta}_{\mathrm{sal}}, \mathcal{H}_{\mathrm{sal}}, Y_{\mathrm{sal}}, \mathsf{obs}_{\mathrm{sal}}, \mathsf{exp}_{\mathrm{sal}}, \perp_{\mathrm{sal}})$ where:

- $\mathbf{\Theta}_{\mathrm{sal}}$ is the space of CNN weight configurations θ (convolutional filters, batch-norm parameters, classifier head);
- $\mathcal{H}_{\mathrm{sal}} = [0,1]^{H \times W}$ is the space of ReLU-activated GradCAM heatmaps, normalized to $[0,1]$;
- Y_{sal} is the space of input-to-output prediction functions $f_\theta : \mathcal{X} \to \mathcal{Y}$;
- $\mathsf{obs}_{\mathrm{sal}}(\theta) = f_\theta$ records the observable prediction behavior;
- $\mathsf{exp}_{\mathrm{sal}}(\theta)$ is the GradCAM heatmap produced by θ on a fixed input image x_0; and
- $(\mathsf{h}_1, \mathsf{h}_2) \in \perp_{\mathrm{sal}}$ when $\mathrm{argmax}_{(i,j)}\ \mathsf{h}_1 \neq \mathrm{argmax}_{(i,j)}\ \mathsf{h}_2$—the two heatmaps place peak activation at different spatial locations.

The Rashomon property. Sanity checks for saliency methods [Adebayo et al., 2018] demonstrate that saliency maps from networks that differ only in their random initialization can diverge substantially while maintaining identical test accuracy. Hooker et al. [2019] further show that saliency-guided feature removal affects models inconsistently across retraining runs. These findings establish the Rashomon property for $\mathcal{S}_{\mathrm{sal}}$: there exist weight configurations θ_1, θ_2 with $f_{\theta_1} \approx f_{\theta_2}$ yet peak GradCAM activation at different spatial locations.

Proposition 25 (Instance 7: Saliency Map Impossibility)**.** *If $\mathcal{S}_{\mathrm{sal}}$ has the Rashomon property, then no saliency map explanation can simultaneously be faithful, stable, and decisive.*

Proof. Direct application of Theorem 6 to $\mathcal{S}_{\mathrm{sal}}$. □

Empirical validation. We validate Proposition 25 by creating 10 perturbed ResNet-18 models on CIFAR-10 ($\sigma = 0.02$ Gaussian weight noise). With **78.8% prediction agreement** (confirming approximate functional equivalence; prediction agreement is 78.8%, reflecting sensitivity of ImageNet-pretrained ResNet-18 to perturbation on CIFAR-10), the **peak flip rate** is **9.6%** [95% CI: 8.8–10.5%]: nearly one in ten model pairs places peak GradCAM activation at a different spatial location despite the same prediction. The mean pairwise spatial IoU (top-20% threshold) is **95.4%** [CI: 95.3–95.5%], confirming high but imperfect overlap. The negative control (identical weights) yields 100% IoU and 0% flip rate. The resolution test (averaged GradCAM heatmaps) yields a mean IoU of **96.9%**—a **+1.5%** improvement over pairwise IoU, confirming that orbit averaging reduces instability. Full results appear in Table 4.

Table 4: GradCAM Saliency Map Instability (ResNet-18, CIFAR-10). Ten functionally equivalent models (pretrained ResNet-18 + Gaussian perturbations, $\sigma = 0.02$). Spatial IoU uses top-20% threshold; peak flip rate counts pairs where argmax differs by >2 pixels. Negative control uses identical weights (model 0 for all 10 slots). Resolution test averages heatmaps; mean IoU vs. average exceeds pairwise IoU. 95% bootstrap CIs in parentheses.

Metric	Positive	Control	Resolution
Prediction agreement	0.788 (0.752, 0.822)	1.000	—
Mean pairwise IoU	0.954 (0.953, 0.955)	1.000 (1.000, 1.000)	0.969 (0.968, 0.971)
Peak flip rate	0.096 (0.088, 0.105)	0.000 (0.000, 0.000)	—

Moderate instability compared to attention. The 9.6% peak flip rate for GradCAM is substantially lower than the 60% argmax flip rate for attention maps (Section 4.2). This disparity reflects the structural constraint imposed by spatial locality: GradCAM operates on 7×7 or 8×8 feature maps where neighboring activations are highly correlated, whereas token-level attention distributes mass across an unstructured sequence. Spatial smoothness constrains the degree to which perturbations

can rearrange saliency, producing a tighter Rashomon set for the peak-location incompatibility relation. Nevertheless, the impossibility still holds: even moderate instability suffices for the Rashomon property, and the 9.6% flip rate is statistically significant with a bootstrap CI excluding zero.

4.8 Instance 8: Token Citation Explanations Under Model Multiplicity

Large language models are increasingly asked to explain their own predictions: "Classify this review as positive or negative and explain which words drove your decision." We formalize such self-explanations as citation-based summaries—template explanations citing the top-k most-attended tokens—and show that they inherit the impossibility of Theorem 6. Recent work on unfaithful chain-of-thought reasoning [Turpin et al., 2024] and measuring faithfulness of LLM explanations [Lanham et al., 2023] motivates the concern that LLM self-explanations are unreliable even for models that agree on predictions.

The explanation system. We instantiate the abstract framework (Definition 1) as the 6-tuple $\mathcal{S}_{\mathrm{llm}} = (\mathbf{\Theta}_{\mathrm{llm}}, \mathcal{H}_{\mathrm{llm}}, Y_{\mathrm{llm}}, \mathsf{obs}_{\mathrm{llm}}, \mathsf{exp}_{\mathrm{llm}}, \perp_{\mathrm{llm}})$ where:

- $\mathbf{\Theta}_{\mathrm{llm}}$ is the space of transformer weight configurations θ (attention heads, feed-forward layers, embeddings);
- $\mathcal{H}_{\mathrm{llm}} = \binom{[T]}{k}$ is the set of size-k subsets of token positions, representing the k tokens cited in the explanation (we use $k = 3$);
- Y_{llm} is the prediction space (e.g., binary sentiment labels);
- $\mathsf{obs}_{\mathrm{llm}}(\theta)$ is the predicted label for input sentence x;
- $\mathsf{exp}_{\mathrm{llm}}(\theta)$ extracts the top-k content tokens by attention rollout importance as the "cited evidence"; and
- $(S_1, S_2) \in \perp_{\mathrm{llm}}$ when the highest-ranked token differs: $\max(S_1) \neq \max(S_2)$ under the attention ordering—i.e., the explanations disagree on the primary evidence token.

The Rashomon property. The attention rollout instability documented in Section 4.2 directly implies the Rashomon property for $\mathcal{S}_{\mathrm{llm}}$: models with identical predictions but different attention distributions cite different tokens as primary evidence. Ye and Durrett [2022] demonstrate empirically that LLM-generated explanations are unreliable, changing substantially under minor prompt or model variations while predictions remain stable. Turpin et al. [2024] show that chain-of-thought explanations can be systematically unfaithful to the model's actual reasoning process.

Proposition 26 (Instance 8: Token Citation Impossibility)**.** *If $\mathcal{S}_{\mathrm{llm}}$ has the Rashomon property, then no token citation explanation method can simultaneously be faithful, stable, and decisive. Token citations are extracted via attention rollout as a proxy for language model self-explanation.*

Proof. Direct application of Theorem 6 to $\mathcal{S}_{\mathrm{llm}}$. □

Empirical validation. We validate Proposition 26 using 10 perturbed DistilBERT models ($\sigma = 0.01$, matching the attention experiment) on 100 sentiment sentences. Despite **100% prediction agreement**, the **explanation flip rate** (fraction of model pairs citing a different primary token) is **34.5%** [95% CI: 33.2–35.9%], and the mean pairwise **citation overlap** (Jaccard similarity of top-3 token sets) is only **0.798** [CI: 0.791–0.806]. The **negative control** (top quartile of sentences by pairwise Jaccard consensus, 26 sentences) achieves perfect citation overlap (Jaccard = 1.000), confirming that instability is concentrated in genuinely ambiguous sentences. The **resolution test** (consensus top-3 from averaged attention rollout) yields Jaccard overlap of **0.857** [CI: 0.842–0.871], a **+5.8%** improvement over pairwise overlap, confirming that orbit averaging reduces citation instability. Full results appear in Table 5.

Implications for regulatory requirements on LLM explanations. The EU AI Act [EU Parliament, 2024] requires "meaningful explanations" for high-risk AI systems, and practitioners increasingly rely on LLM self-explanations to satisfy this requirement. Proposition 26 and the 34.5% flip rate demonstrate that such explanations are structurally unreliable: retraining (or fine-tuning) the same architecture on the same data can change which tokens the model cites as evidence, even when

Table 5: Token Citation Instability (Attention-Based Citation). 10 perturbed DistilBERT models ($\sigma = 0.01$), 100 sentences. "Citation overlap" = Jaccard similarity of top-3 cited tokens. "Flip rate" = fraction of model pairs where the #1 cited token differs. 95% CIs via bootstrap.

Metric	Value	95% CI
Prediction agreement	100.0%	—
Positive test (all 100 sentences)		
Citation overlap (Jaccard)	0.798	[0.791, 0.806]
Explanation flip rate (#1 token)	0.345	[0.332, 0.359]
Negative control (26 easy sentences: top quartile by Jaccard consensus)		
Citation overlap (Jaccard)	1.000	[1.000, 1.000]
Flip rate	0.315	[0.289, 0.343]
Resolution test (consensus top-3 vs individual top-3)		
Consensus overlap (Jaccard)	0.857	[0.842, 0.871]

the prediction itself is unchanged. This is not a failure of any particular prompting strategy or explanation method; it is a mathematical consequence of underspecification. Regulators should require that LLM explanations be accompanied by stability audits—e.g., reporting citation overlap across multiple model instances—rather than treating a single model's self-explanation as ground truth.

4.9 Instance: Mechanistic Interpretability (Circuit Discovery)

Mechanistic interpretability (MI) has emerged as a central paradigm for AI safety research, aiming to reverse-engineer neural networks into human-understandable circuits [Elhage et al., 2022, Conmy et al., 2023, Anthropic, 2025]. We instantiate Theorem 6 for circuit discovery, where the fundamental obstruction is that functionally equivalent networks admit multiple valid circuit decompositions.

The explanation system. Define the 6-tuple $\mathcal{S}_{\text{circuit}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows.

- $\mathbf{\Theta}$ is the set of neural network weight configurations (architecture plus trained parameters).
- $\mathcal{H}$ is the set of *circuit decompositions*: subgraphs of the computational graph together with an interpretation of which components implement which sub-computations (e.g., "heads 1.3 and 2.1 form an induction circuit").
- Y is the set of input–output functions: each $y \in Y$ is the function $f_\theta : \mathcal{X} \to \mathcal{Y}$ computed by the network on the domain of interest.
- $\mathsf{obs} : \mathbf{\Theta} \to Y$ maps each weight configuration θ to the function it computes.
- $\mathsf{exp} : \mathbf{\Theta} \to \mathcal{H}$ maps each network to its circuit decomposition—the internal computational mechanism identified by a circuit discovery method.
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ is the *component-attribution incompatibility*: $(h, h') \in \perp$ if h and h' attribute the same computation to disjoint or contradictory subsets of network components (e.g., h says the computation is implemented by heads in layer 2, while h' says it is implemented by heads in layer 4).

The Rashomon property: circuit multiplicity. Meloux et al. [2025] provide striking empirical evidence that the Rashomon property holds for circuit discovery. For a simple XOR task, they identified **85 distinct valid circuits** with *zero circuit error* (each circuit perfectly reproduces the network's behavior on the task). Moreover, each circuit admitted an average of **535.8 valid interpretations**—different assignments of computational roles to components, all equally faithful to the network's input–output behavior.

Complementary evidence comes from Bricken et al. [2025], who show that sparse autoencoders trained on the *same* data learn *different* features depending on the random seed, demonstrating that even the feature-level decomposition underlying circuit identification is not unique.

Proposition 27 (Circuit Rashomon Property)**.** *$\mathcal{S}_{\text{circuit}}$ has the Rashomon property: there exist networks $\theta_1, \theta_2 \in \mathbf{\Theta}$ with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ (identical I/O behavior) yet $(\mathsf{exp}(\theta_1), \mathsf{exp}(\theta_2)) \in \perp$ (incompatible circuit decompositions).*

Proof. Consider any pair of distinct valid circuits from the 85 found by Meloux et al. [2025]: by construction, both achieve zero circuit error (same I/O behavior on the task), yet they attribute the computation to different subsets of network components. □

Instance: mechanistic interpretability impossibility.

Theorem 28 (Mechanistic Interpretability Impossibility)**.** *The explanation system $\mathcal{S}_{\text{circuit}}$ has the Rashomon property. Therefore, by Theorem 6, no circuit explanation method* exp *can simultaneously be:*

1. ***Faithful**: $\mathsf{exp}(\theta)$ reflects the actual internal circuitry of network θ;*
2. ***Stable**: $\mathsf{exp}(\theta_1) = \mathsf{exp}(\theta_2)$ whenever $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ (functionally equivalent networks receive the same circuit explanation); and*
3. ***Decisive**: exp commits to a single circuit decomposition (rather than reporting the space of valid circuits).*

Proof. By Proposition 27, $\mathcal{S}_{\text{circuit}}$ has the Rashomon property. Apply Theorem 6. □

This is verified in Lean 4 as `mech_interp_impossibility` in `MechInterpInstance.lean` (three type axioms: `CircuitConfig`, `CircuitExplanation`, `CircuitObservable`, plus the `circuitSystem` axiom encoding the Rashomon witness).

Critical framing: what this does and does not say. This result does *not* undermine mechanistic interpretability—it clarifies what MI can and cannot achieve. Circuit discovery methods such as activation patching [Conmy et al., 2023], sparse probing [Elhage et al., 2022], and circuit tracing [Anthropic, 2025] remain powerful tools for understanding neural network internals. What the impossibility establishes is that the *search for THE circuit*—a single, definitive decomposition—is provably constrained: any method that commits to a unique answer must sacrifice either faithfulness or stability.

The constructive path forward is to *characterize the space of valid circuits*, not to search for a single canonical one. This is precisely the direction that leading MI work is already moving toward: Anthropic [2025] report confidence levels rather than single circuits, and Conmy et al. [2023] frame automated circuit discovery as search over a structured space.

Resolution: circuit equivalence classes. The uniform resolution of Section 5 specializes to mechanistic interpretability as follows. Let $G = [C]$ denote the equivalence class of circuits that reproduce the same I/O behavior. The G-invariant explanation aggregates over all valid circuits:

$$\mathsf{exp}^*(\theta) \;=\; \{C \in \mathcal{H} : C \text{ is a valid circuit for } f_\theta\},$$

the set of all circuit decompositions consistent with the network's behavior. This map is **faithful in expectation** (it captures all structure shared across valid circuits), **stable** by construction (functionally equivalent networks have the same set of valid circuits), but **not decisive**: it returns a set rather than a single circuit. The unresolvable queries—which specific components "really" implement a computation when multiple valid assignments exist—correspond precisely to the undirected edges in the causal discovery analogy (Section 4.5).

Reporting the full equivalence class is the principled response to the impossibility, trading decisiveness for faithfulness and stability, exactly as predicted by Theorem 6. This reframes the MI research agenda: the goal is not to find the one true circuit, but to characterize which circuit features are *invariant* across all valid decompositions (the "skeleton" of the circuit space) and which are decomposition-dependent (the "reversible edges").

Table 6: Empirical validation of the universal explanation impossibility across nine explanation types. Seven types have direct experimental results; mechanistic interpretability and causal discovery are validated via classical results. All instability measures have 95% bootstrap confidence intervals excluding zero. Prediction equivalence confirms models are functionally identical.

Instance	Instability metric	Value	Prediction equiv.
Attribution (GBDT)	SHAP flip rate	33%	>99%
Attention (DistilBERT)	Argmax flip rate	60.0%	100%
Counterfactual (XGBoost)	Direction flip rate	23.5%	ΔAUC < 0.03
Concept probe (MLP)	$1 - \lvert\cos\rvert$	0.90	97.2%
Model selection (XGBoost)	Best-model flip rate	80.0%	ΔAUC < 0.02
Saliency map (ResNet-18)	Peak flip rate	9.6%	78.8%[†]
Token citation (DistilBERT)	Citation flip rate	34.5%	100%[‡]
Mech. interp. (circuits)	*85 valid circuits, Meloux et al., 2025*		
Causal discovery	*Classical (Verma & Pearl, 1991); Rashomon Derived*		

[†]Prediction agreement is 78.8%, reflecting sensitivity of ImageNet-pretrained ResNet-18 to perturbation on CIFAR-10.

[‡]Token citation extracted via attention rollout as a proxy; four of twenty attribution models achieved 72–78% held-out accuracy; all twenty included.

Table 7: The three escape routes from the explanation trilemma.

Drop	What you lose	What you gain	Tools
Faithful	May contradict model	Stable + decisive	Fixed rules, random circuit
Stable	Changes on retrain	Faithful + decisive	SHAP, GradCAM, single circuit
Decisive	Ties, uncertainty	Faithful + stable	DASH, CPDAG, ensemble, equiv. classes

5 The Universal Resolution

The impossibility of Theorem 6 is tight: one of the three desiderata must be sacrificed. In this section we show that a *uniform* resolution exists whenever the Rashomon property arises from a symmetry group acting on the hypothesis space. The resolution sacrifices decisiveness and achieves faithfulness in expectation together with exact stability.

5.1 G-Invariant Explanation Systems

Let G be a group acting on the hypothesis space $\mathcal{H}$. The action is *observable-preserving* if for every $g \in G$ and $h \in \mathcal{H}$,

$$\mathsf{obs}(g \cdot h) \;=\; \mathsf{obs}(h),$$

i.e., the group permutes hypotheses within observe-equivalence classes (fibers). We say that G *acts transitively on observe-fibers* if for every pair h, h' with $\theta(h) = \theta(h')$, there exists $g \in G$ such that $g \cdot h = h'$.

Definition 29 (G-Invariant Explanation Map)**.** *An explanation map* exp *is* G-invariant *if for every* $g \in G$ *and* $h \in \mathcal{H}$*,*

$$\mathsf{exp}(g \cdot h) \;=\; \mathsf{exp}(h).$$

When G is finite and Y admits averaging, the canonical G-invariant map is the *orbit average*:

$$\overline{\mathsf{exp}}(h) \;=\; \frac{1}{|G|} \sum_{g \in G} \mathsf{obs}(g \cdot h).$$

Proposition 30 (G-Invariance Implies Stability)**.** *If* G *acts transitively on observe-fibers and* exp *is* G*-invariant, then* exp *is stable.*

Proof. Let $h, h' \in \mathcal{H}$ with $\theta(h) = \theta(h')$. By transitivity, there exists $g \in G$ with $g \cdot h = h'$. By G-invariance, $\mathsf{exp}(h') = \mathsf{exp}(g \cdot h) = \mathsf{exp}(h)$. (Formalized as `gInvariant_stable` in `UniversalResolution.lean`.) □

Table 8: Instance-specific resolutions of the Universal Explanation Impossibility. Each row identifies the symmetry group, the concrete resolution, and the property sacrificed.

Instance	Symmetry group G	Resolution	Sacrificed
Attribution	Collinear-feature permutations	DASH (ensemble SHAP averaging)	Decisive ranking
Attention	Weight-space symmetries	Averaged attention rollout	Single top token
Counterfactual	Decision-boundary variation	Rashomon-robust recourse	Single counterfactual
Concept probe	Representation rotation	Averaged CAV / subspace	Single direction
Causal discovery	DAG reversal within Markov class	CPDAG reporting	Fully oriented graph
Model selection	Model permutation in Rashomon set	Ensemble / Rashomon set report	Single best model
Saliency map	Weight-space symmetries	Averaged GradCAM heatmap	Single peak location
Token citation	Weight-space symmetries	Averaged attention rollout	Single cited token
Mech. interpretability	Circuit permutations	Circuit equivalence class	Single circuit

Proposition 31 (Design Space Dichotomy)**.** *If the explanation system has the Rashomon property, then no stable explanation map can simultaneously be faithful and decisive.*

Proof. Immediate from Theorem 6: stability, faithfulness, and decisiveness are jointly contradictory under the Rashomon property. (Formalized as `universal_design_space_dichotomy` in `UniversalDesignSpace.lean`.) □

Corollary 32 (Pareto Frontier)**.** *For any explanation system with the Rashomon property, achievable explanation maps fall into exactly two families:*

- ***Family A*** *(faithful + decisive, unstable): individual-model explanations that faithfully report each hypothesis but vary across equivalent hypotheses.*
- ***Family B*** *(stable + faithful in expectation, indecisive): G-invariant aggregations that are constant on observe-fibers but produce ties for queries where the fiber disagrees.*

No explanation map lies outside these two families.

5.2 Instance-Specific Resolutions

Table 8 summarizes how the G-invariant resolution instantiates across the nine domains of Section 4.

5.3 DASH as the Prototype Resolution

DASH (**D**iversified **A**ggregation of **SH**AP) is the G-invariant resolution for the feature attribution instance (Section 4.1). It averages SHAP values over an ensemble of models retrained from independent random seeds, achieving $\mathbb{E}[\phi_j] = \mathbb{E}[\phi_k]$ for any pair of symmetric features j, k (*consensus equity*). This orbit average is stable by Proposition 30 and faithful in expectation: the expected explanation matches the expected observation across the Rashomon set. The cost is indecisiveness—DASH reports ties for feature pairs whose rankings disagree across the ensemble. The companion paper proves that DASH is Pareto-optimal: no other stable attribution method can achieve higher expected faithfulness.

5.4 Meta-Theorem: Orbit Averaging

Theorem 33 (Orbit-Averaged Resolution)**.** *Let $\mathcal{S}$ be an explanation system whose Rashomon property arises from a finite group G acting on $\mathcal{H}$ and acting transitively on observe-fibers. Then the orbit-averaged map $\overline{\mathsf{exp}}(h) = \frac{1}{|G|}\sum_{g\in G} \mathsf{obs}(g \cdot h)$ is:*

1. *well-defined (independent of orbit representative),*
2. *stable (Proposition 30),*
3. *faithful in expectation: $\mathbb{E}_{g\sim \mathrm{Unif}(G)}[\mathsf{obs}(g \cdot h)] = \overline{\mathsf{exp}}(h)$, and*
4. *Pareto-optimal among stable maps: no stable map achieves strictly higher pointwise faithfulness on every fiber.*

The orbit-averaged resolution is constructive: for any specific instance, the group G and its action are determined by the domain structure, and the averaging can be computed (or approximated by Monte Carlo sampling over the Rashomon set). Theorem 33 is the abstract counterpart of the DASH optimality result and is formalized in `UniversalResolution.lean`.

6 Ubiquity

Is the Universal Explanation Impossibility a pathological corner case, or a generic phenomenon? In this section we show that it is the latter: the Rashomon property—and hence the impossibility—holds whenever the model has more parameters than observational constraints, a condition satisfied by virtually all modern machine learning systems.

6.1 Generic Underspecification

Proposition 34 (Dimensional Underspecification)**.** *Let $\mathcal{H}$ be a smooth hypothesis space of dimension $\dim(\mathcal{H})$ and let $\mathsf{obs} : \mathcal{H} \to Y$ be a smooth observation map into a space of dimension $\dim(Y)$. If $\dim(\mathcal{H}) > \dim(Y)$, then for generic $y \in Y$, the fiber $\mathsf{obs}^{-1}(y)$ has positive dimension $\geq \dim(\mathcal{H}) - \dim(Y)$.*

This is a direct consequence of the preimage theorem in differential topology. Whenever the fiber has positive dimension, the explanation map obs is generically non-constant on fibers—different hypotheses in the same fiber honestly report different explanations—and the Rashomon property holds. (Formalized as `generic_underspecification` in `Ubiquity.lean`; the bridge from fiber non-degeneracy to the impossibility is `fiber_nondegeneracy_implies_impossibility`.)

6.2 Neural Network Dimensional Argument

Modern neural networks are massively overparameterized. A ResNet-50 has $P \approx 25{,}600{,}000$ parameters and, for ImageNet classification, $k = 1{,}000$ output classes. By Proposition 34, the fiber of any fixed prediction has dimension at least $P - k = 24{,}999{,}000$. On such a high-dimensional manifold, any continuous explanation map (attention weights, gradient saliency, internal activations) is generically non-constant: there exist models h, h' on the same fiber with $\mathsf{obs}(h)_j > \mathsf{obs}(h)_k$ and $\mathsf{obs}(h')_j < \mathsf{obs}(h')_k$ for any pair of input features j, k with equal marginal distributions.

Corollary 35 (Coin-Flip Instability)**.** *For any pair of symmetric features j, k in an overparameterized network trained to zero loss, the probability (over retraining seeds) that a faithful explanation ranks j above k is exactly $1/2$.*

This is the neural-network instantiation of the symmetric Bayes result formalized in `SymmetricBayes.lean`: the symmetry of the fiber under feature permutation forces uniform marginals on the ranking.

6.3 Prevalence

The dimensional condition $\dim(\mathcal{H}) > \dim(Y)$ is the norm, not the exception. Large language models (billions of parameters, vocabulary-sized output), diffusion models, graph neural networks, random forests with hundreds of trees, and Lasso regression with more features than observations all satisfy it. Even linear regression with collinear features produces a positive-dimensional fiber (the collinear subspace). D'Amour et al. [2022] document this across NLP, computer vision, and medical imaging, finding that underspecification—and hence explanation instability—is ubiquitous in deployed systems. The only escape is a model class so constrained that $\dim(\mathcal{H}) \leq \dim(Y)$, which excludes essentially all modern architectures of practical interest.

6.4 Analogy: Arrow's Impossibility Theorem

The structure of our result parallels Arrow's impossibility theorem in social choice theory [Arrow, 1951]. Arrow showed that no voting rule satisfying unanimity, independence of irrelevant alternatives, and non-dictatorship exists for three or more candidates—a universal impossibility that unified disparate paradoxes in voting theory. Nipkow [2009] later formalized Arrow's theorem in

Isabelle/HOL. Our Universal Explanation Impossibility plays an analogous role for explainability: it unifies nine domain-specific impossibilities under one structural cause (the Rashomon property) and one four-line proof, and the resolution (orbit averaging) is the XAI counterpart of accepting partial orderings in social choice. (The full impossibility is formalized as `ubiquity_impossibility` in `Ubiquity.lean`.)

7 Lean Formalization

The Universal Explanation Impossibility and all nine instances are mechanically verified in Lean 4 [de Moura and Ullrich, 2021] using Mathlib [The mathlib Community, 2020]. Table 9 summarizes the formalization.

Table 9: Summary of the Lean 4 formalization.

Component	Count	Notes
Files	73	All in `UniversalImpossibility/`
Theorems & lemmas	346	Verified by `grep -c`
Axioms	72	Domain-specific; core theorem uses 0
`sorrys`	0	Zero admitted goals
Core impossibility	1 theorem	Zero axiom dependencies
GBDT instance	4 axioms	Surjectivity, first-mover, split structure
DASH resolution	6 axioms	+ cross-group symmetry
Causal instance	3 axioms	CPDAG Markov equivalence (Rashomon derived)
Neural net instance	2 axioms	Overparameterization + symmetry
Fairness instance	2 axioms	Unequal base rates + equal accuracy

The core impossibility (`universal_explanation_impossibility` in `UniversalImpossibility/Universal.lean`) depends on zero axioms beyond the Rashomon property, which is stated as a hypothesis rather than an axiom. Each instance introduces the minimum axioms needed to verify that its particular explanation system satisfies the Rashomon property.

The Lean proof of Theorem 6 is four lines, mirroring the paper proof exactly. The formalization also includes:

- The G-invariant resolution (Section 5): `Corollary.lean`, `DesignSpace.lean`;
- Quantitative bounds for GBDT (`Ratio.lean`), Lasso (`Lasso.lean`), and neural networks (`NeuralNet.lean`);
- The fairness instance (`FairnessAudit.lean`) and causal instance (`CausalDiscovery.lean`).

8 Discussion: What Explainability Can and Cannot Do

8.1 The Ceiling of Explainability

The Universal Explanation Impossibility (Theorem 6) establishes a *ceiling*, not a criticism of any particular method. No axiomatic redesign of SHAP, LIME, attention maps, or counterfactual generators can simultaneously deliver faithfulness, stability, and decisiveness whenever the underlying inference problem is underspecified. The ceiling is structural: it follows from the Rashomon property alone, independent of the explanation formalism. An analogy to physics is apt: the speed of light constrains spacecraft engineering but does not prevent useful space travel. Likewise, the impossibility constrains what practitioners may promise about explanations but does not prevent useful explainability—it redirects effort toward the achievable frontier characterized by the G-invariant resolution (Section 5).

8.2 Regulatory Analysis: SR 11-7

OCC SR 11-7 [Office of the Comptroller of the Currency, 2011] establishes three pillars for model validation in banking that map directly onto the trilemma. Section III.A (*Conceptual Soundness*) requires that "the model's theoretical construct, … assumptions, and limitations should be assessed"—this demands *faithfulness*: the explanation must reflect the model's actual structure and assumptions. Section III.B (*Outcomes Analysis*) requires "comparing the model's outputs with the corresponding realized outcomes" across "model versions"—this demands *stability*: the explanation must be consistent across model versions that produce equivalent outcomes. Section IV (*Effective Challenge*) requires "critical analysis by objective, informed parties" yielding "specific findings"—this demands *decisiveness*: the challenge must produce specific, actionable findings, not hedged non-answers.

These three requirements, as stated, cannot be simultaneously satisfied for any banking model with the Rashomon property—i.e., any model where multiple parameter configurations achieve equivalent predictive performance. For models trained with bagging, boosting, or stochastic gradient descent, the Rashomon property is generic (Section 5). Theorem 6 applies directly.

Compliant model documentation under the trilemma should: (a) state which property is prioritized for each model use case; (b) use ensemble explanations (DASH) for the stable + faithful tradeoff, documenting where decisiveness is sacrificed; (c) report explanation uncertainty bounds (confidence intervals on feature importance across the Rashomon set) alongside point estimates; and (d) document the Rashomon set size and the resulting explanation variance as part of the model risk assessment. Effective challenge (Section IV) is then directed at modeling choices—why this Rashomon set? why this loss threshold?—rather than at explanation artifacts that are mathematically underdetermined.

The same analysis applies to EU AI Act Article 13 [EU Parliament, 2024] (transparency) and NIST AI RMF MAP 1.5 [National Institute of Standards and Technology, 2023] (interpretability). In each case, the regulatory language implicitly assumes all three properties are jointly achievable; our theorem shows they are not. Regulations should be reformulated as explicit tradeoffs: specify which property is required, which may be relaxed, and what resolution method is acceptable.

8.3 What Explainability *Can* Do

The G-invariant resolution (Section 5) is constructive, not nihilistic. It tells the practitioner exactly which desiderata can be jointly satisfied and which must be relaxed. DASH for feature attribution, CPDAGs for causal discovery [Verma and Pearl, 1991], and ensemble concept probes for concept-based explanations [Kim et al., 2018] each sacrifice decisiveness to gain stability—and each is a concrete instance of the abstract resolution. In practice, this means reporting confidence intervals over the Rashomon set [Fisher et al., 2019, Semenova et al., 2022] rather than point estimates: a feature ranking should come with a stability annotation indicating which comparisons are robust to retraining and which are not. The impossibility does not say "explanations are useless"; it says "decisive explanations are unreliable under underspecification, but partially decisive explanations with calibrated uncertainty are both achievable and informative."

8.4 The "Isn't This Obvious?" Defense

A natural reaction is that practitioners already know explanations are unstable —so what is new? The contribution is fivefold. First, we make explicit that nine domains (attribution, attention, counterfactuals, concept probes, causal discovery, model selection, saliency maps, token citation, mechanistic interpretability) share *one* mechanism—the Rashomon property—rather than nine unrelated pathologies. Second, the Lean 4 formalization (Section 7: 73 files, 346 theorems, 0 `sorry`) eliminates logical gaps in the formal proofs that informal arguments inevitably leave. Third, the empirical experiments (Section 4) show quantifiable severity: 33% attention flip rates, measurable counterfactual and concept instability on standard benchmarks. Fourth, the resolution via G-invariant maps is constructive and actionable, not merely diagnostic. Fifth, Lean-verified tightness theorems (Proposition 9) prove the impossibility is non-trivial: each pair of properties *is* simultaneously achievable, so the theorem is not an artifact of overly strong definitions—it identifies the exact boundary between the possible and the impossible. Arrow's impossibility theorem [Arrow, 1951] was arguably "obvious" once stated—every committee chair knows that voting rules are imperfect. Yet its for-

malization created the field of social choice theory, precisely because it replaced intuition with a theorem that could be built upon.

8.5 The "All Instances Are Trivial" Defense

A reviewer might object that each individual instance is already known: attention instability was documented by Jain and Wallace [2019], Markov equivalence is classical [Verma and Pearl, 1991], fairness impossibility was proved by Chouldechova [2017], and SHAP instability was observed by D'Amour et al. [2022] and Krishna et al. [2022]. The contribution is not any single instance but the *meta-theorem*: the same four-line proof (Theorem 6) closes all nine cases under a single structural condition. Before this work, a practitioner encountering attention instability might switch to SHAP, then to counterfactual explanations, then to concept probes—each time hoping the new method escapes the problem. Our theorem shows that no such escape exists: the impossibility is structural (it follows from the Rashomon property), not method-specific. Recognizing this unifies the practitioner's response: rather than debugging each method individually, one diagnoses the Rashomon property and applies the G-invariant resolution uniformly.

8.6 Implications Beyond Machine Learning

The Universal Explanation Impossibility is not specific to machine learning. We briefly sketch five domains where the theorem applies directly.

Philosophy of science: formalizing Quine–Duhem. Quine [Quine, 1951] and Duhem [Duhem, 1954] argued that empirical evidence underdetermines theory choice. Van Fraassen [van Fraassen, 1980] elevated this to a central argument for constructive empiricism. Despite a century of philosophical discussion, a core consequence of the thesis has not, to our knowledge, been formalized as a mathematical theorem with a precise boundary condition. Theorem 6 provides that formalization: the configuration space corresponds to theories, the observable space to data, the Rashomon property to the existence of empirically equivalent theories with incompatible explanations, and the necessity result (Proposition 10) supplies the exact boundary that philosophical formulations lack. Okasha [Okasha, 2011] connected theory choice to Arrow's impossibility in social choice; our result provides the complementary connection for underdetermination of explanation.

Neuroscience: degeneracy as the Rashomon property. Marder and Goaillard [Marder and Goaillard, 2006] documented that crustacean stomatogastric ganglia producing identical motor patterns are built from qualitatively different ionic conductance profiles. Edelman and Gally [Edelman and Gally, 2001] formalized this as *degeneracy*: structurally distinct elements performing the same function. Degeneracy is precisely the Rashomon property in our framework, and the impossibility theorem applies: no mechanistic account of neural circuit function can simultaneously be faithful to the specific circuit, stable across degenerate circuits, and decisive about which ionic mechanism is primary.

Economics: partial identification. Manski [Manski, 2003] developed partial identification theory for settings where observational data do not pin down a unique model. The identified set corresponds to an observe-fiber in our framework; partial identification is the economic instantiation of the Rashomon property. The G-invariant resolution corresponds to reporting bounds on the identified set rather than a point estimate.

Medical diagnosis: differential diagnosis as resolution. When multiple conditions produce the same symptoms, diagnosis faces the trilemma. Differential diagnosis—the standard medical practice of listing all conditions consistent with the symptoms—is precisely G-invariant resolution. It sacrifices decisiveness (commitment to a single diagnosis) to gain faithfulness and stability, exactly as Theorem 6 predicts.

8.7 Limitations and Open Questions

We identify five limitations and directions for future work.

1. **Axiomatized Rashomon properties.** The Rashomon property for each instance is axiomatized—verified to hold under stated assumptions—rather than derived from first principles of the learning algorithm. Deriving the Rashomon property from, e.g., the loss landscape geometry of gradient descent would strengthen the result.
2. **Weight perturbation vs. true retraining.** The attention instability experiments (Section 4) use weight perturbation within the zero-loss manifold rather than full retraining from independent random seeds. While this faithfully models the Rashomon set, the distribution over perturbations may differ from the distribution induced by stochastic gradient descent.
3. **Non-attribution quantitative bounds.** For the attribution instance, the companion paper provides sharp quantitative bounds (divergence rates, ensemble size requirements, DASH optimality). Analogous quantitative bounds for the counterfactual, concept, and fairness instances remain open.
4. **Decisiveness as a practical cost.** The G-invariant resolution sacrifices decisiveness: it reports ties where the data are ambiguous. Whether downstream decision-makers can tolerate partial rankings or tied causal structures—and how to present such outputs effectively—is an empirical question we do not address here.
5. **Fairness instance not experimentally validated.** The fairness audit impossibility (Instance 6, formalized in `FairnessAudit.lean`) is proved but not empirically validated in this paper. Chouldechova [2017] provides the original empirical context; replicating it within our framework is left to future work.

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